

LOGGING SENSOR DATA In MS EXCEL Through MATLAB GUI

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The measurement and analysis of process parameters such as temperature, flow, conductivity, speed, viscosity and stress play a crucial role in providing information about the process/system under consideration. Parameter values are logged into a computer (using a process called data acquisition) and acquired data is analysed either in online or offline mode. Time and frequency domain analysis of the data is then carried out to provide various information about the system, such as its mathematical model, static and dynamic characteristics, fault identification and so on.

This project presents a MATLAB graphical user interface (GUI)-based approach to save real-time process data obtained from a temperature sensor (LM35) in MS Excel. The GUI allows the user to graphically view the temperature variation at the end of sensor data acquisition. A screenshot of the GUI is shown in Fig. 1.

Hardware and construction

Circuit diagram for interfacing the temperature sensor with Arduino is shown in Fig. 2.

Arduino Uno. Arduino Uno is an AVR ATmega328P microcontroller (MCU)-based development board having six analogue and 14 digital I/O pins. The MCU has 32kB ISP flash memory, 2kB RAM and 1kB EEPROM. The board allows serial communication via UART, SPI and I2C. The MCU can operate at a clock frequency of 16MHz. In this project, analogue pin A0 of Arduino is used to read the

output voltage of LM35.

LM35. LM35 is a precision IC temperature sensor, which provides output voltage linearly proportional to temperature (in degree Celsius). It exhibits typical accuracies of $\pm 1/4^\circ\text{C}$ at room temperature and $\pm 3/4^\circ\text{C}$ over a full temperature range from -55 to $+150^\circ\text{C}$. The sensor provides a sensitivity of $10\text{mV}/^\circ\text{C}$. In this project, Vout pin (pin 2) of LM35 IC is connected to analogue pin A0 of Arduino. Pins 1 and 3 of LM35 are connected to +5V and Gnd of Arduino Uno, respectively.

Software

The application program is developed using R2014a version of MATLAB. Note that, the program will not work with earlier versions of MATLAB. Additionally, Legacy MATLAB and Simulink Support for Arduino package must also be installed. The procedure for installation of ArduinoIO package is given below.

1. After downloading ArduinoIO, extract the compressed folder named ArduinoIO
2. From ArduinoIO folder, copy pde folder and paste it in C:\Program Files(X86)\Arduino\Examples
3. Open Arduino IDE
4. Open code from File→Examples→pde→adiaoies
5. Connect Arduino Uno to the PC
6. Upload adiaoies code to Arduino Uno
7. Copy the entire extracted ArduinoIO folder to a folder in My Documents
8. Open MATLAB 2014
9. Open install_arduino.m file present in ArduinoIO folder in My Documents
10. Run install_arduino.m file

The code will install and save the path of Arduino support package

The PC must also have MS Excel or OpenOffice installed. The program can write sensor data only on MS Excel using `xlswrite()` MATLAB function.

After correctly setting up the path for the package, open MATLAB source code file called `excel_log.m`

Edit COM port number {in `a = arduino('COM7')` line} with

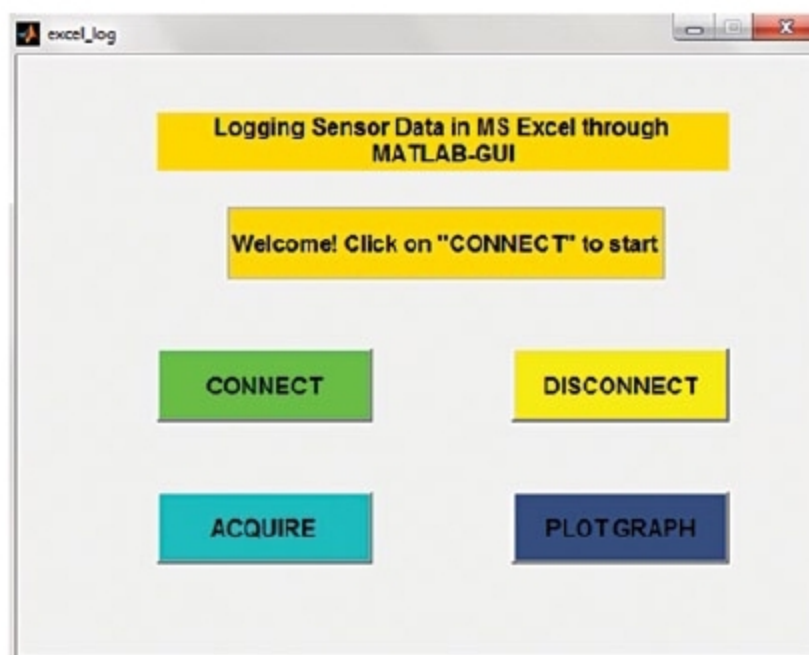


Fig. 1: Screenshot of the GUI

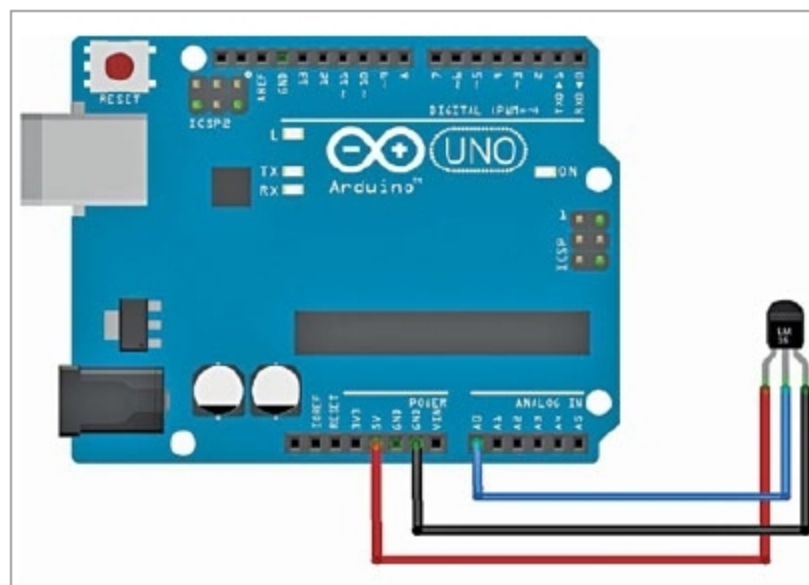


Fig. 2: Circuit diagram of temperature sensor interfaced with Arduino